

An Auxiliary Rehabilitation Device for Patients with Muscle Tremors and Stiffness

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Teacher Sign:

Two handwritten signatures in black ink. The first signature is a stylized, cursive 'R' followed by a horizontal line. The second signature is a more complex, cursive script with multiple loops and a long horizontal stroke extending to the right.

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1. Introduction

Patients coping with conditions such as Parkinson's often face challenges with irregular tremors and finger stiffness, resulting in diminished normal finger functionality. Ongoing efforts are being made to enhance rehabilitation and provide daily assistance to individuals facing medical conditions that impact their hand function.

From a neurorehabilitation point of view, the performance of functional tasks in a highly intensive, repetitive manner is beneficial to the patient ^{[1][2][3][4][5]}. Furthermore, robots enable more independent training with lower supervision, allowing the patient to perform intensive therapy for longer ^[6]. In terms of daily life assistance, the use of robotic devices, namely exoskeletons, could greatly improve the independence of their users ^[7].

The robotic device is comprised of three main components: 1) rehabilitation/execution component; 2) actuation/driving component; 3) control component. See an overview of the components in Table 1.1 below ^[8].

Table 1.1 Overview of components of robotic device

Execution component	Actuation component	Control component
• Exoskeleton devices • End-effector devices • Hand orthoses	• Pneumatic actuation • Linear actuation • Cable-driven • Gear-motor actuation	• Position control with PID • Model-based computed torque control • Admittance control • Force sensor-based control • Impedance control

As rehabilitation/execution component, exoskeletons is more popular and helpful in daily assistance of the hand rehabilitation^[8] because home rehabilitation has been shown to be beneficial for functional and psychological performance ^[9] and would ideally lead to shorter yet more effective therapy regimes.

Among medical applications, soft robots for rehabilitation are increasingly being employed owing to their inherent safety, light weight, small size and increased potential for portability

and efficacy, in contrast to conventional rigid rehabilitation ^[9]. Soft robot hand rehabilitation gloves have evolved over years to provide a variety of features and functionalities for hand therapy and rehabilitation.

The existing rehabilitation equipment could be summarized into two different categories based on their driving method and actuator types.

The first category uses motor drive and linear actuator. M Sandison, et al, developed HandMATE (Hand Movement Assisting Therapy Exoskeleton) ^[10], a wearable motorized hand exoskeleton for home-based movement therapy following stroke. Each finger is powered by a linear actuator which provides flexion and extension assistance. The device is shown in Figure 1.1 below. Motors are used to drive the actuators. A cable-driven linkage system was used inside the exoskeleton to pull the fingers ^[10]. This can implement concise control, but the wire in cable-driven system is easily got slackened.



Figure 1.1 HandMATE device

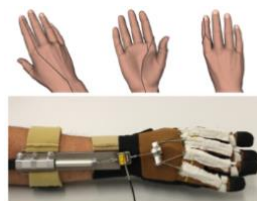


Figure 1.2 Linkage system

Shown in Figure 1.2, exotendon routes for achieving different movement patterns illustrated for index finger (top) and whole-hand movement pattern implemented with single-actuator exotendon network (bottom).

The second category uses pump drive, fluid (water/pneumatic) actuator. P. Polygerinos et al. ^[11], developed a robotic glove utilizing soft actuators that induce specific flexion, twisting, and extending trajectories under fluid pressurization. These soft actuators can support the range of motion of individual fingers with significant force when pressurized and low impedance when un-actuated, which can make the soft robotic glove to carry out functional grasping. A pump is used as source of energy to drive the actuators. The actuators are soft

tubes with fiber reinforcements. The tubes are fixed on the fingers to pull/push them. A control hardware system was designed and included fluidic pressure sensors in line with the actuators and a closed-loop controller to regulate the pressure. The illustration of the device is shown in Figure 1.3.

In our auxiliary rehabilitation device, the actuator employs the Extensible Pneumatic Muscle (EPM) structure within the finger module, comprising soft tubes reinforced with fiber. Additionally, the finger module, which incorporates flex sensors and pressure sensors, effectively regulates the inflation and deflation of pneumatic joints, facilitating the flexion and extension of the patient's fingers during rehabilitation training. The

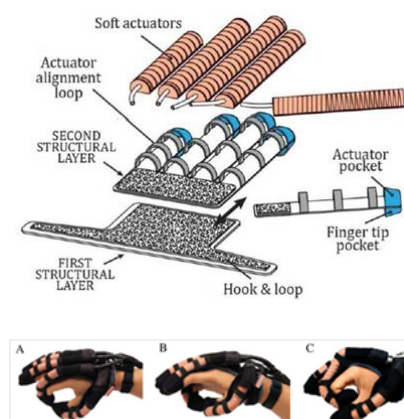


Figure 1.3: soft robotic glove assembly.

flex sensors are designed to track flexion angles, whereas the pressure sensors evaluate appropriate pressure levels to ensure effective risk management. Moreover, the device assists in the classification of Parkinson's disease diagnoses by acquiring finger motion data through an accelerometer. The device can also be used for assisted living by actively controlling tremor muscles.

This design incorporates several improvements compared to previous devices:

1. Finger extension is regulated by an auxiliary pump, enabling slow finger movements and thus significantly improving user comfort and functional variety.
2. Equipped with an accelerometer, the device provides a feature for scaling the rating of finger/hand tremors for Parkinson's patients.
3. The device offers a 'life assistance' mode for daily life, which can prevent accidental clicks by Parkinson's patients due to finger tremors before the mouse cursor reaches the intended target.

4. This design incorporates a mobile app, allowing users to effortlessly and swiftly connect to rehabilitation equipment. It leverages the portability of both mobile phones and rehabilitation equipment, enhancing convenience of the device.

2. Theoretical Research

The device includes a finger module containing Extensible Pneumatic Muscle (EPM) structures, which serve as the main components responsible for executing flexion and extension of the fingers. This chapter will detail the theoretical design of the Extensible Pneumatic Muscle (EPM).

2.1 Extensible Pneumatic Muscle Dimension

As mentioned above, the Extensible Pneumatic Muscle (EPM) is utilized as an actuating component in this design. It is affixed to the fingers to assist with their flexion and extension. Theoretically, it is essential to determine the dimensions of the tube with bellows (EPM components) to ensure it is sufficiently long to accommodate the full range of finger movement. This section will calculate the dimensions of the tube with bellows.

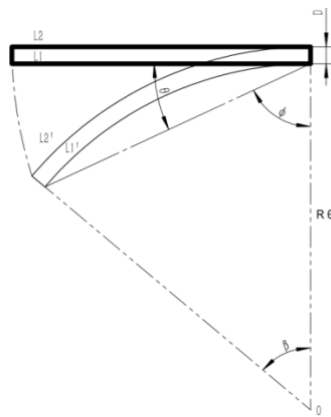


Figure 2.2: Model of the bellows changing as the finger bends

Assume the length of the finger-driven bellows: L_1 represents the length of the lower side, and L_2 represents the length of the upper side. When the fingers are not bent, $L_1 = L_2 = L_0$. The diameter of the bellows is denoted as D . The finger flexion angle, θ , is defined as the angle between the fingertip and the root of the finger. Here, the finger's curvature radius is

approximated as R_θ , with O as the center, and β as the angle subtended by the fingertip, the root of the finger, and the center of the curvature.

Because: $\frac{\beta}{2} + \emptyset = 90^\circ$, $\theta + \emptyset = 90^\circ$, we get:

$$\beta = 2\theta \quad (1)$$

After the finger is bent, the length L_1 of the lower edge remains unchanged, then: $L_1' = L_1$

Also, because: $L_1 = R_\theta * \beta$, then we get:

$$R_\theta = \frac{L_0}{2\theta} \quad (2)$$

The length L_2 of the upper edge of the bellows is unchanged to L_2' , we have:

$$L_2' = (R_\theta + D) * \beta = \left(\frac{L_0}{2\theta} + D\right) * 2\theta = L_0 + 2D\theta \quad (3)$$

When the finger is flexed to a θ degree angle, the resulting change in length of L_2 is as follows:

$$\Delta L_2 = L_2' - L_2 = 2D\theta \quad (4)$$

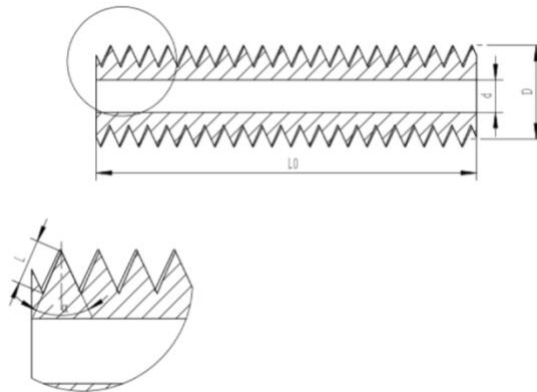


Figure 2.3: Structure of the bellows

In Figure 2.3, l represents the length of the hypotenuse of the corrugated protrusion, α is the angle of inclination of the corrugation, d is the inner diameter of the bellows, and D is the outer diameter of the bellows. The length l can then be calculated as:

$$l = \frac{D-d}{2} * \frac{1}{\cos \frac{\alpha}{2}} = \frac{D-d}{2 \cos \frac{\alpha}{2}} \quad (5)$$

Suppose the number of corrugations or folds of the bellows is n . The change in the length of the bellows when fully expanded after inflation, compared to the original length, needs to be greater than the change in length (ΔL_2) required when the fingers are bent, that is:

$$2nl - L_0 \geq \Delta L_2 \quad (6)$$

The length of the bellows can be expressed as:

$$L_0 = n * 2 * \frac{D - d}{2} * \tan \frac{\alpha}{2} = n * (D - d) * \tan \frac{\alpha}{2}$$

From this, we deduce that:

$$n = \frac{L_0}{D - d} * \frac{1}{\tan \frac{\alpha}{2}} \quad (7)$$

Substituting (4)(5)(7) into (6), we get:

$$2nl - L_0 \geq \Delta L_2$$

Further calculation yields:

$$2 * \frac{L_0}{D - d} * \frac{1}{\tan \frac{\alpha}{2}} * \frac{D - d}{2 \cos \frac{\alpha}{2}} - L_0 \geq 2D\theta$$

$$\Rightarrow \frac{L_0}{\sin \frac{\alpha}{2}} - L_0 \geq 2D\theta$$

$$\Rightarrow \sin \frac{\alpha}{2} \leq \frac{L_0}{L_0 + 2D\theta} \quad (8)$$

Taking the maximum value $\theta_{\max}$ of the finger flexion angle θ , equation (8) becomes:

$$\sin \frac{\alpha}{2} \leq \frac{L_0}{L_0 + 2D\theta_{\max}}$$

Hence:

$$\alpha \leq 2 \sin^{-1} \frac{L_0}{L_0 + 2D\theta_{\max}} \quad (9)$$

Based on equation (9), we can select an appropriate length and diameter D of the bellows and then design the inclination angle α of the bellows.

2.2 Movement Angle Assigning

In this design, the finger is simplified into two components: the metacarpus and the phalanges (distal, middle, and proximal), as illustrated in Figure 2.4. The movement of a single finger is actuated by two finger-driving structures based on the Extensible Pneumatic Muscle (EPM) structure, hereafter referred to as the finger movement structure. This section aims to theoretically calculate the required flexion angles of each EPM muscle structure when the finger transitions from one position to another.

To position a finger at the desired location, the two finger movement structures must coordinate and flex by specific angles, respectively. A mathematical model is established here to determine the angles by which the two movement structures should flex, transitioning from the finger's initial position to the desired position. The 'desired position' is denoted by a point in the coordinate system, while the initial position of the finger is along the y-axis.



Figure 2.4: Anatomy structure of the finger

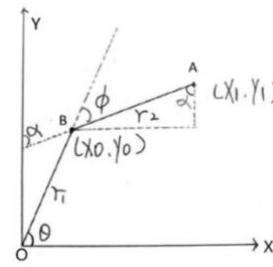


Figure 2.5: Finger joints positions in relation with finger flexion angles

As shown in Figure 2.5, $A(x_1, y_1)$ is the position of fingertip (distal phalange), $B(x_0, y_0)$ is the position of the MCP (see Figure 2.4); O is the joint of metacarpus and palm; r_1 is length of metacarpus, r_2 is the length of phalange; θ is the angle of point B ; ϕ is the angle between metacarpus and phalange.

As illustrated in the graph above, the position of point $A(x_1, y_1)$ can be described as:

$$x_1 = x_0 + r_2 \cdot \sin \alpha \quad (1)$$

$$y_1 = y_0 + r_2 \cdot \cos \alpha \quad (2)$$

Also:

$$\alpha = (90^\circ - \theta) + \phi \quad (3)$$

$$x_0 = r_1 \cdot \cos \theta \quad (4)$$

$$y_0 = r_1 \cdot \sin \theta \quad (5)$$

Substituting (3)(4)(5) into equation (1) and (2), we get:

$$x_1 = x_0 + r_2 \cdot \sin(90^\circ - \theta + \phi) = x_0 + r_2 \cdot \cos(\phi - \theta) = r_1 \cdot \cos \theta + r_2 \cdot \cos(\phi - \theta)$$

$$y_1 = y_0 + r_2 \cdot \cos(90^\circ - \theta + \phi) = y_0 - r_2 \cdot \sin(\phi - \theta) = r_1 \cdot \sin \theta - r_2 \cdot \sin(\phi - \theta)$$

That is:

$$x_1 = r_1 \cdot \cos \theta + r_2 \cdot \cos(\phi - \theta) \quad (6)$$

$$y_1 = r_1 \cdot \sin \theta - r_2 \cdot \sin(\phi - \theta) \quad (7)$$

In which x_1, y_1, r_1, r_2 are given.

Our purpose is to find solution for ϕ, θ .

Because:

$$\cos(\phi - \theta) = \cos \phi \cdot \cos \theta + \sin \phi \cdot \sin \theta$$

$$\sin(\phi - \theta) = \sin \phi \cdot \cos \theta - \cos \phi \cdot \sin \theta$$

Substituting them into (6) (7), we get:

$$x_1 = r_1 \cdot \cos \theta + r_2 \cdot (\cos \phi \cdot \cos \theta + \sin \phi \cdot \sin \theta)$$

$$y_1 = r_1 \cdot \sin \theta - r_2 \cdot (\sin \phi \cdot \cos \theta - \cos \phi \cdot \sin \theta)$$

Simplifying them, we get:

$$x_1 = (r_1 + r_2 \cdot \cos \phi) \cdot \cos \theta + r_2 \cdot \sin \phi \cdot \sin \theta \quad (8)$$

$$y_1 = (r_1 + r_2 \cdot \cos \phi) \cdot \sin \theta - r_2 \cdot \sin \phi \cdot \cos \theta \quad (9)$$

Next, by squaring both equations and adding them together, we get:

$$x_1^2 + y_1^2 = (r_1 + r_2 \cdot \cos \phi)^2 + r_2^2 \cdot \sin^2 \phi$$

This simplifies to:

$$x_1^2 + y_1^2 = r_1^2 + 2r_1r_2 \cos \phi + r_2^2$$

With further calculation, it yields:

$$\cos \emptyset = \frac{x_1^2 + y_1^2 - r_1^2 - r_2^2}{2r_1 r_2}$$

So, we get the solution of $\emptyset$:

$$\emptyset = \arccos \left(\frac{x_1^2 + y_1^2 - r_1^2 - r_2^2}{2r_1 r_2} \right) \quad (10)$$

Next, we calculate the solution for θ .

Rewrite equation (8) as below:

$$x_1 = b \cdot \cos \theta + c \cdot \sin \theta \quad (11)$$

Where:

$$b = r_1 + r_2 \cdot \cos \emptyset \quad (12)$$

$$c = r_2 \cdot \sin \emptyset \quad (13)$$

Divide both sides of (11) by $\sqrt{b^2 + c^2}$ to obtain:

$$\frac{x_1}{\sqrt{b^2 + c^2}} = \frac{b}{\sqrt{b^2 + c^2}} \cdot \cos \theta + \frac{c}{\sqrt{b^2 + c^2}} \cdot \sin \theta$$

Let

$$d = \sqrt{b^2 + c^2} \quad (14)$$

And

$$\beta = \arctan \left(\frac{c}{b} \right) \quad (15)$$

Then equation (11) can be expressed as:

$$\frac{x_1}{d} = \cos(\theta - \beta)$$

Therefore:

$$\theta = \beta + \arccos \left(\frac{x_1}{d} \right) \quad (16)$$

Substituting (14) (15) into (16), we arrive at the solution for θ :

$$\theta = \arctan \left(\frac{r_2 \cdot \sin \emptyset}{r_1 + r_2 \cdot \cos \emptyset} \right) + \arccos \left(\frac{x_1}{r_1^2 + r_2^2 + 2r_1 \cdot r_2 \cdot \cos \emptyset} \right) \quad (17)$$

Thus, the solutions for φ and θ are given in equations (10) and (17).

3. Materials and Methods

3.1 Design Overview

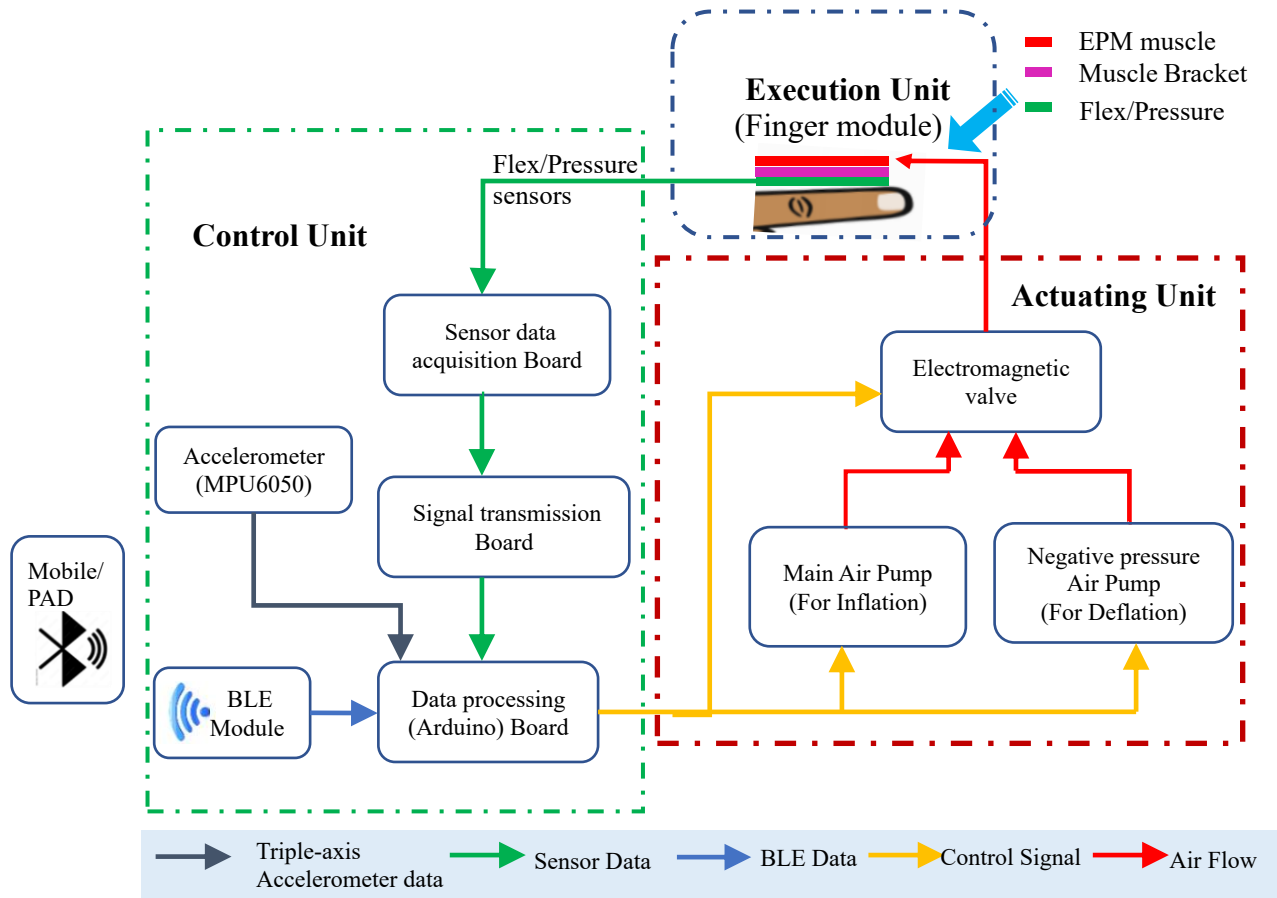


Figure 3.1 System Block Diagram of the Rehabilitation Device

The device comprises of three main units: Control unit, Actuating Unit, and Execution unit.

- 1) The Control unit consists of Sensor data acquisition Board, Signal transmission Board, Accelerometer, Bluetooth module and Data processing Board (Arduino Mega control development board). This unit is responsible for collecting and processing sensors data.
- 2) The actuating unit consists of the main air pressure pump, electromagnetic valves, and a negative-pressure air pump. According to the commands from the Arduino board, the actuating unit controls the airflow to regulate the inflation and deflation of pneumatic joints.

- 3) The execution unit is the finger module, which comprises Extensible Pneumatic Muscle (EPM) structures, muscle brackets, flex sensors, and pressure sensors. The finger module actively controls the flexion and extension of the fingers during finger motion control.
- 4) The Mobile/PAD can communicate with the rehabilitation device by Bluetooth protocol.

3.2 Mechanical Design

Among these units of the device, the Execution Unit (Finger module, refer to Figure 3.2) is the essential self-designed mechanical component of this device, which is assembled from Extensible Pneumatic Muscle (EPM) Structure, Muscle Bracket, Flex sensors and Pressure sensors. It is one of the critical components of the Rehabilitation device.

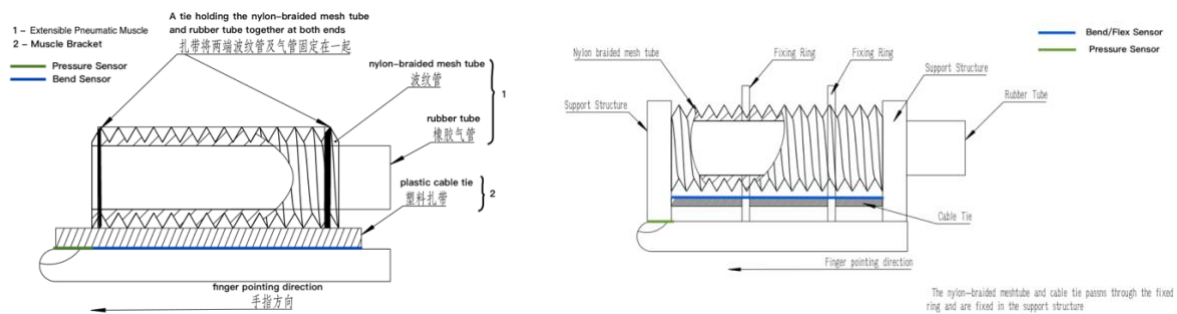


Figure 3.2 a) Finger Module V 1.0

b) Finger Module V 2.0

This design underwent five iterations of MB design, correspondingly iterating the entire auxiliary rehabilitation device design twice. The table below provides descriptions of the respective iterations.

Table 3.1 Design iteration description

Version of Finger Module	Version of Muscle Bracket	Number of finger module for each finger	Number of sensors for each finger	Space between EPM and Muscle Bracket	Other Distinctive features
1.0	0	1	2	EPM is tightly affixed to Muscle Bracket	Conceptual prototype
	1.0	1	2	EPM is tightly affixed to Muscle Bracket	Testing prototype
	1.5	2	4	EPM is tightly	Close to true

				affixed to Muscle Bracket	finger structure and fine control
2.0	2.0	2	4	EPM is lifted from Muscle Bracket	Lift the muscle and get bigger force arm
	2.5	2	4	EPM is lifted from Muscle Bracket	Lighter muscle bracket weight

This chapter will describe the mechanical design of Finger Module in detail.

3.2.1 Extensible Pneumatic Muscle (EPM)

The Extensible Pneumatic Muscle (EPM), also known as a pneumatic artificial muscle, is an actuator designed to emulate the contraction of human muscles through pneumatic principles. EPM typically consists of a nylon-braided mesh tube enveloping a rubber tube, secured with cable ties and closed off with a tracheal plug. The operating mechanism of an EPM is predicated on the inflation and deflation of this flexible tube which is usually crafted from nylon or rubber. Introducing compressed air or fluid into the tube induces its expansion and contraction, thereby allowing the EPM to perform work similar to that of a biological muscle, offering advantages in applications requiring smooth and natural motion.

In this project, the EPM muscle structure was modified to be more suitable for device (refer to Figure 3.2). Instead of the pneumatic muscle structure contracting, the modified EPM muscle structure elongates when pressurized with air. With the appropriate design, this can achieve the effect of the EPM muscle structure's flexion after being introduced with compressed air and maintain that posture until depressurized.

3.2.2 Muscle Bracket

In this design, the Muscle Bracket (MB) is the fixed bracket of the EPM muscle in the finger module, as shown in Figure 3.3. The MB is essential because it limits the sideways expansion of the EPM muscle and transforms it into a flexion motion. The principle is: as the EPM muscle elongates during inflation, the fixed length of the MB creates tension, resulting in a flexion motion. Five different generations or versions of the MB were developed, each

building on improvements from the previous one. The next section will explore the distinct designs of these five MB generations.

Muscle Bracket V0

This version is the conceptual prototype for the whole design. In this design, the EPM muscle structure is directly affixed to a plastic cable tie using adhesive; Sensors are attached beneath the cable tie at the point of contact with the finger.

Muscle Bracket V1.0

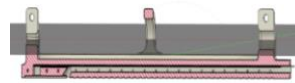


Figure 3.3 a) Bottom view of MB 1.0

b) Side View of MB V1.0

The V1.0 MB initially follows with the concept of having one Finger Module for each finger. The entire MB features a rectangular design to ensure a close fit between the muscle structure and the MB during movement. The muscle structure passed through a central fixing ring, clamped at both ends, with adhesive filling any gaps. To reduce resistance, transverse grooves beneath the MB were designed, folding during the downward flexion process.

Each Finger module section contains two sensors: a Flex sensor detecting finger flexion and a Pressure sensor assessing fingertip pressure for pain detection. The MB intentionally provides space for sensor placement, with a hollowed central portion accommodating wiring, and the Pressure sensor positioned within a groove on the front segment (shown as Figure3.3).

Muscle Bracket V1.5

In MB V1.5, to closely replicate the anatomical structure of human fingers and enhance fine hand control, each finger adopts a design consisting of two nearly identical MB V1.0 components. The connection between these two MB units is established through specialized muscle bracket structures.

Each EPM muscle structure was connected to its respective air tube and sensors as shown in Figure 3.5.



Figure 3.4 Muscle Bracket Version 1.5



Figure 3.5 Finger module Version 1.5

Physical product side look

Muscle Bracket V2.0

MB V2.0 includes support structures at both ends, a fixing ring in the middle, and a cable tie (refer to Figure 3.2 b).

A key feature of this version is that the EPM muscle is not tightly affixed to the MB but is lifted, increasing the lever arm for muscle flexion and reducing the flexion force. Flex and pressure sensors are positioned through holes in the support structure and fixing ring.

Advantages of MB Version 2.0:

- 1) Addresses insufficient power from the air pump by increasing the arm of force of the EPM muscle structure.
- 2) Resolves the connection issue between EPM muscle and MB by elevating the muscle structure away from the Muscle Bracket.

Two grooves in the support structures (see Figure 3.6) securely lock the ends of EPM muscle structure, while the middle section is fastened with fixing rings and cable ties (Figure 3.2 b). Gaps between fixing rings allow force transfer during muscle structure movement and provide space for extension. This design eliminates the need for adhesive, enclosing the muscle structure between fixing rings and supporting structures, effectively securing it.

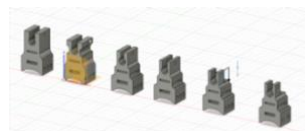


Figure 3.6 MB Version 2.0 support structure

As shown in Figure 3.6, several minor changes and updates were made on the support structures based on the outcomes, the one on the far right and the fourth from the left were finally used. The Figure 3.7 below shows the Muscle Bracket version 2.0.

As shown in Figure 3.6, several minor changes and updates were made on the support structures based on the outcomes, the one on the far right and the fourth from the left were finally used. The Figure 3.7 below shows the Muscle Bracket version 2.0.



Figure 3.7 MB Version 2.0 Prototype



Figure 3.8 MB Version 2.5 Prototype

Muscle Bracket V2.5

The differences of V2.5 design from the V2.0 are: 1) the support structures at both ends of the Bracket are simpler and lighter; 2) The fixing ring in the middle is thinner; 3) Some hollow rectangular bars are added between the two support structures on the tie belt below to limit the movement of the fixing ring as shown in Figure 3.8. Figure 3.10 is the final design of Rehabilitation device version 2.0 with MB version 2.5.



Figure 3.9 Rehabilitation Device V1.0
(Finger module V1.0, Covering MB V0, 1.0, 1.5)

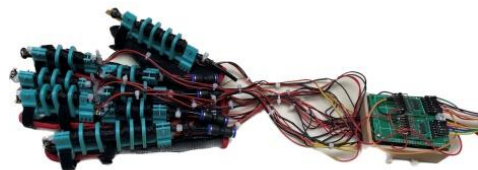


Figure 3.10 Rehabilitation Device V2.0
(Finger module V2.0, Covering MB V2, 2.5)

3.3 Electronic System Design

The electrical system in the Rehabilitation device design allocates in the three units: Control Unit, Actuating Unit and Execution Unit. The Figure 3.11 shows the device electronic system connection diagram.

- 1) Control Unit includes five parts:
 - a) Sensor Data Acquisition Board with a voltage conversion circuit: This board serves to intricately collect data from Flex or Pressure sensors.
 - b) Sensor Data Transmission Board, which is an I2C board (used in device V 0) or an AD extender (used in device version V2.0). The primary purpose of the Sensor

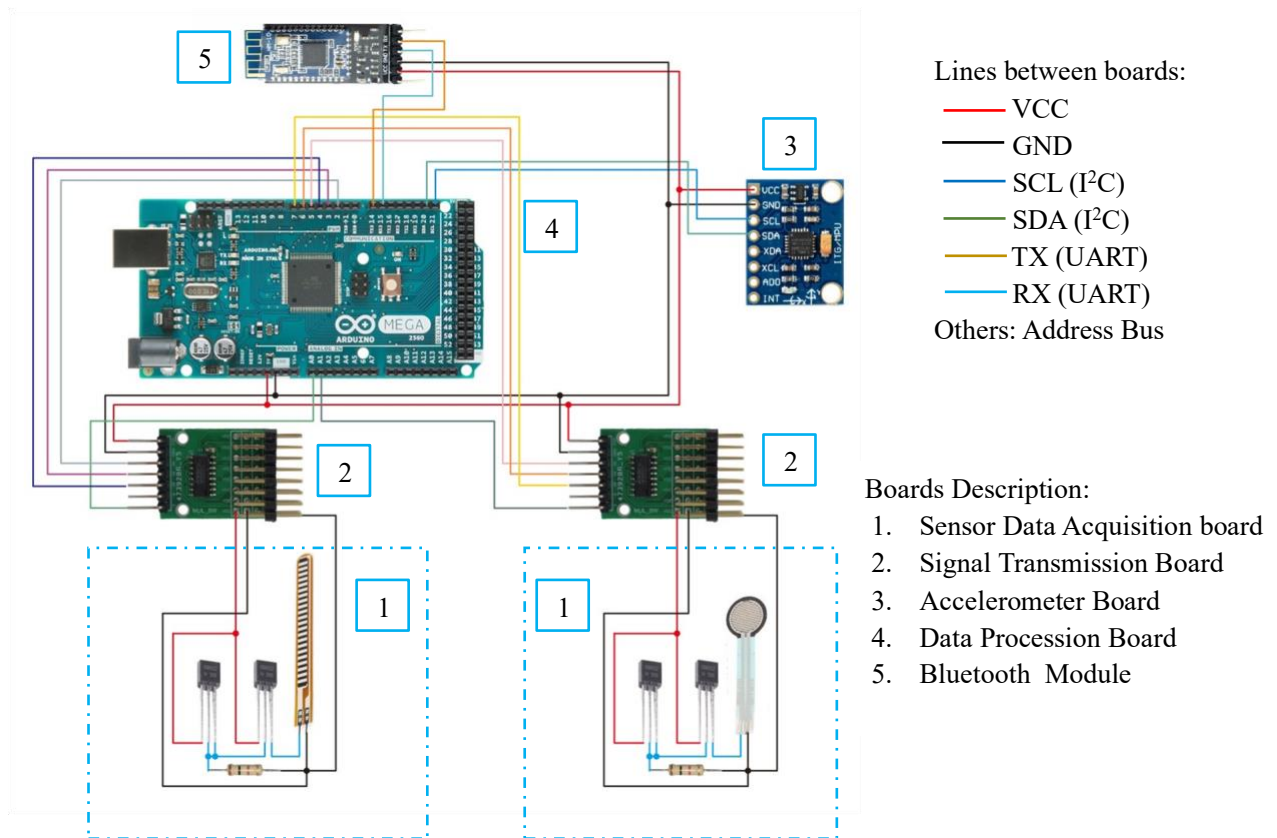


Figure 3.11 Electronic System Design Diagram

Data Transmission Board is to transmit sensor data to the processing board.

- c) Accelerometer Board: Equipped with an MPU6050 sensor, keeps track of the finger tremor duration and rate. This functionality is particularly useful for patient monitoring at home.
- d) Data Processing Board, featuring an Arduino Mega (microcontroller): This board receives and processes sensor data and controls valves and pumps.
- e) Bluetooth module: A peripheral of Arduino board, enables connection between mobile/PAD and Bluetooth module.

2) Actuating Unit

- a) main air pump
- b) negative pressure air pump
- c) MOS controlled electromagnetic valve

This unit regulates the airflow into the EPM muscle to control the inflation and deflation of the fingers.

3) Execution Unit

a) Finger module: Flex sensor and Pressure sensor.

The execution unit achieves the rehabilitation training by the inflation and deflation of the EPM muscle, resulting in finger's flexion and extension.

The whole rehabilitation device functions based on the data collected from the sensors and pumps air flow control, the Control unit will be described in the electrical design section.

3.3.1 Sensor Data Acquisition Board

The Sensor Data Acquisition unit includes:

- 1) B1 - a 5V power supply.
- 2) RV1 - a slide potentiometer, which corresponds to the sensor in this design – a pressure sensor^{[15][17]} or flex sensor^{[16][17]}.
- 3) Q1 and Q2 - two S9012 PNP transistors^[13]
- 4) R2 - a 100 Ω color-coded resistor^[14].

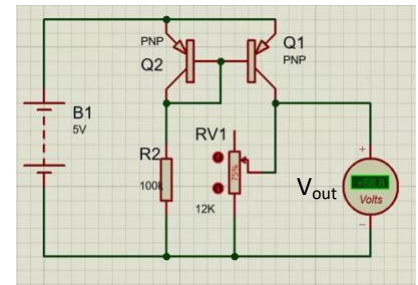


Figure 3.12 Sensor Data Acquisition Unit Schematic

In the circuit: Q1 and Q2 are the model, so: $I_{B1}=I_{B2}$; β is the magnification of PNP.

$$I_{R2} = I_{C2} + I_{B2} + I_{B1} = \beta I_{B2} + I_{B2} + I_{B2} = (\beta + 2) I_{B2}$$

$$I_{VR1} = I_{C1} = \beta I_{B1} = \beta \frac{I_{R2}}{\beta + 2} ; \quad V_{out} = I_{VR1} * R_{VR1} = \beta \frac{I_{R2}}{\beta + 2} * R_{VR1}$$

V_{out} changes as the resistance value of VR1 changes.

A single finger module of the device is equipped with a flex sensor and a pressure sensor. In this rehabilitation device, the thumb and little finger each utilize one finger module unit, while each of the three other fingers employs two finger module units. Consequently, the entire device uses a total of 8 groups, comprising 16 sensors, and the circuitry is ultimately

organized into 16 circuits similar to those illustrated in Figure 3.12 – each circuit group corresponding to one sensor. The voltage values of the 16 signals will be collected through the AD expansion board / I²C board and then passed to the Arduino data processing board.

3.3.2 Sensor Data transmission board

Two data transmission boards were used in this design. The I²C protocol communication board was used in finger module V1.0. To improve the transmission speed, the AD transmission board was used in finger module V2.0.

I²C Communication Board (used in finger module V1.0)

Every four sets of sensors will be implemented with an AD board consisting of ADS1115 for data collection. The data will be transmitted via the I²C protocol to the PCA548A main I²C circuit board, which connecting four ADS115 expansion boards. Ultimately, the data from sixteen sets of sensors will be transmitted to the Arduino Mega control board via this main I²C communication board. See Figure 3.13.

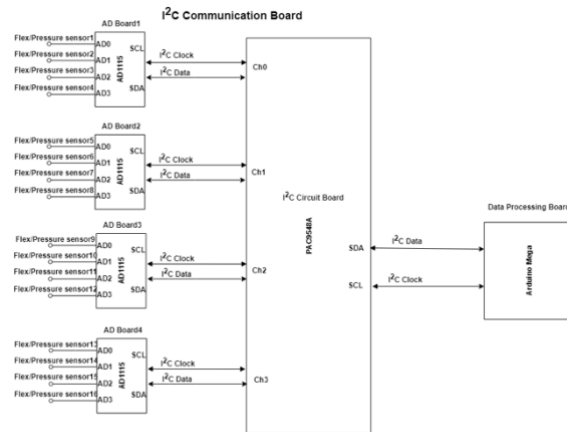


Figure 3.13 I²C Communication Board Diagram

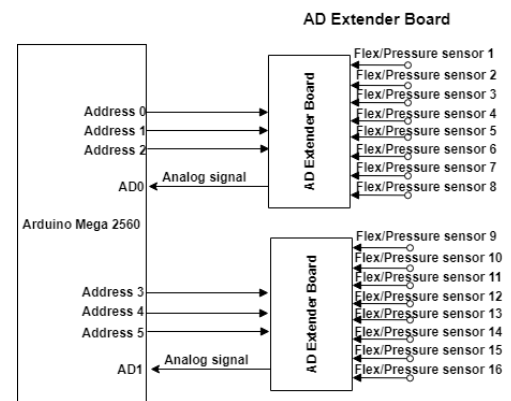


Figure 3.14 AD extender board

A significant challenge with the I²C board is the slow signal transmission speed. Initially, the 16 sets of data were divided into four groups and connected to separate ADS1115 modules.

Subsequently, these four ADS1115 modules were connected to the PCA9548A.

This sequential process consumed a considerable amount of time.

AD Extender board (used in finger module V2.0)

Each eight sensors are connected to an AD extender board. The 16 sensors are divided into two groups, each served by its own AD extender board for data acquisition. Finally, all sensor data is transmitted to the Data Processing Board (Arduino Mega). See Figure 3.14.

The board is suitable for use in analog or digital 8:1 multiplexer application. The switch features three digital select inputs (A0, A1 and A2), eight independent inputs (Y0-Y7), a common output (Z) and a digital enable input (E). When E is HIGH, the switches are turned off. Inputs include clamp diodes.

3.3.3 Accelerometer

This accelerometer is primarily assembled around the MPU6050, featuring a triple-axis MEMS gyroscope. The gyroscope encompasses X-, Y-, and Z-Axis angular rate sensors, with a user-programmable full-scale range of ± 250 , ± 500 , ± 1000 , and $\pm 2000^\circ/\text{sec}$. Simultaneously, the MPU6050 incorporates a triple-axis MEMS accelerometer with various capabilities, including a digital-output system and programmable full-scale range options of $\pm 2g$, $\pm 4g$, $\pm 8g$, and $\pm 16g$. This device aids in Parkinson's diagnostic classification.

3.3.4 Data Processing Board

The Arduino Mega 2560^[12] was selected as the microcontroller board in this design due to its efficiency. It has 54 digital input/output pins (of which 15 can be used as PWM outputs), 16 analog inputs, 4 UARTs (hardware serial ports) which fulfills the need of a large amount of pins in this project. The development environment of Arduino is also relatively friendly, which facilitates the combination of many different types of sensors which will be quite a lot in my project.

In this rehabilitation device, the Arduino Mega board is responsible for:

- 1) Acquisition and processing the sensors (pressure sensor, flex sensor, accelerometer) data
- 2) controlling the on or off of air pumps and open or close of magnetic valves

- 3) Analyzing the past finger flexion data of patient to determine the rehabilitation status
- 4) Based on the accelerometer's data, aids the device in Parkinson's diagnostic classification
- 5) Controlling the finger movement for daily activities (life assistance).

3.3.5 Bluetooth Module

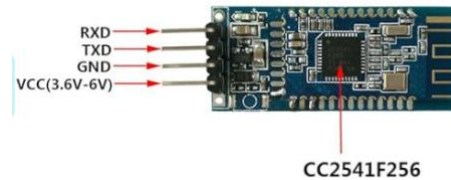


Figure 3.15 HM-10 Bluetooth 4.0 module

The control unit uses the DSD TECH HM-10 Bluetooth 4.0 BLE iBeacon UART Module, which is the Bluetooth module for peripherals of Arduino. The user connects with the rehabilitation device with the mobile phone through the Bluetooth protocol of this module.

The details of this module:

- 1) Working voltage 3.6 V to 6V, and default Baud rate of 9600.
- 2) 4pin Base Board: Through this base board, leads to VCC, GND, TX, RX. It can be very convenient to connect to the Arduino board.

3.4 Software Design

3.4.1 Software Diagram

The Auxiliary Rehabilitation device has function modes: Rehabilitation training, Diagnosis and life assistance. Additionally, the design is portable, and it includes an UI which can be downloaded in the Mobile phone or PAD. The following section will describe the software diagrams respectively for the different modes.

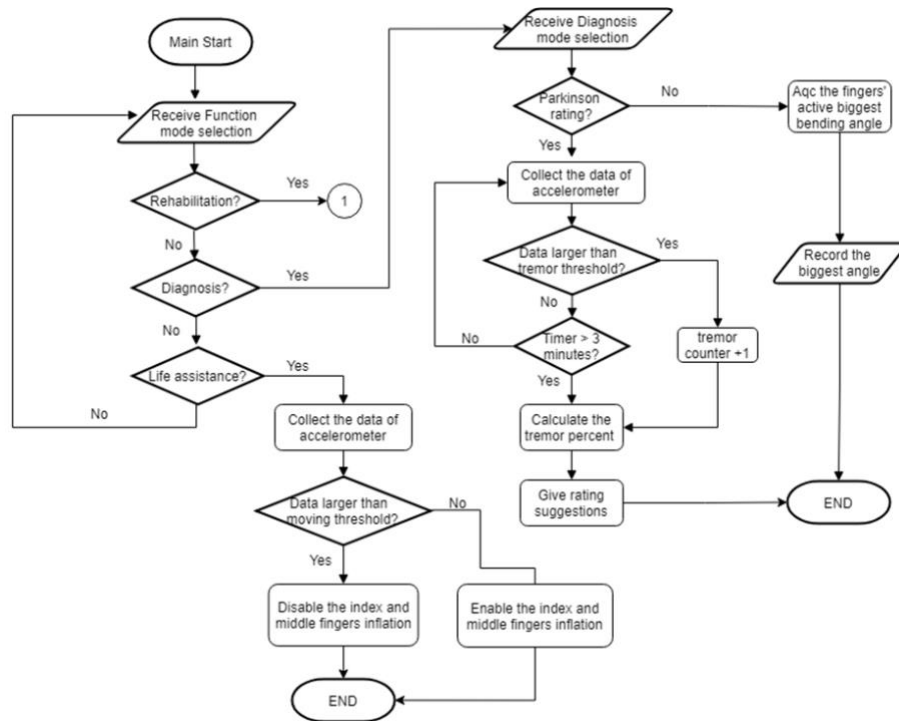


Figure 3.16 (a) Software main function Diagram

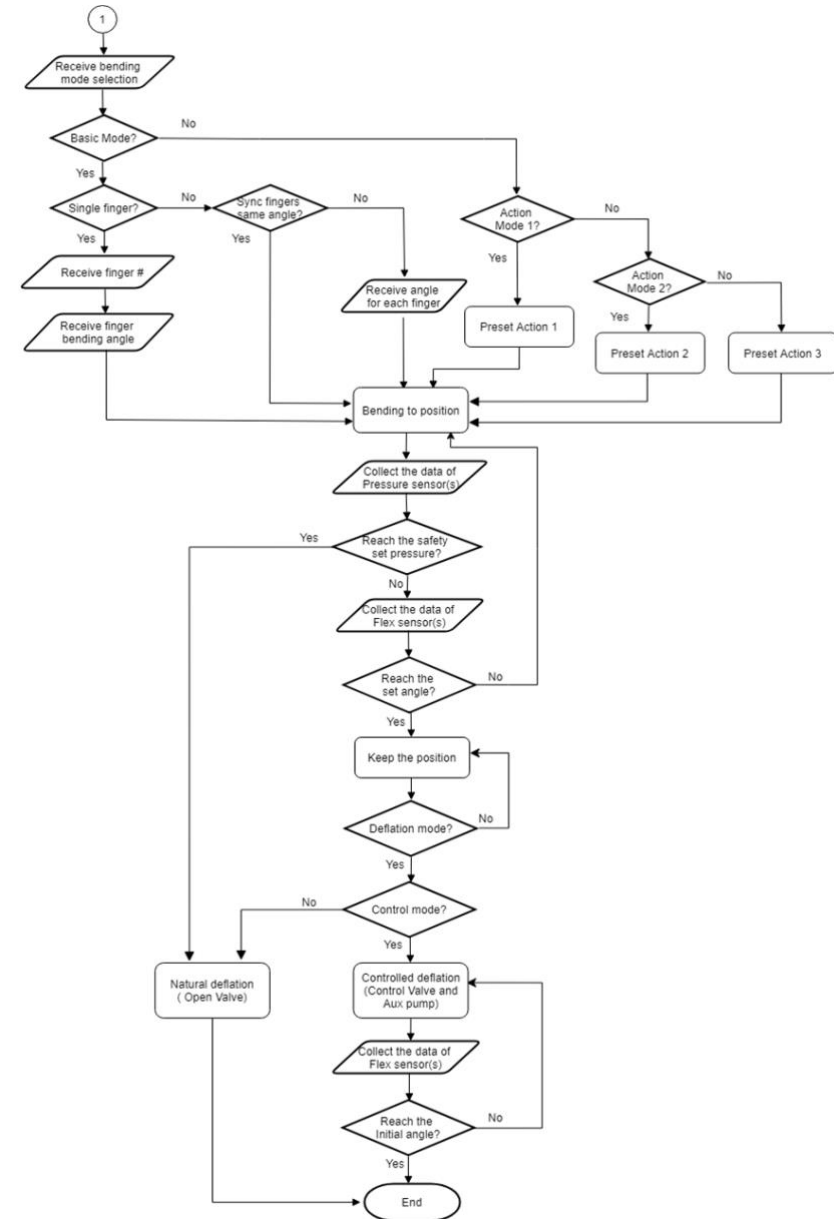


Figure 3.16 (b) Rehabilitation training function Diagram

3.4.2 Software procedures

1) Rehabilitation function: it is used for the Rehabilitation training of tremor fingers.

The device has two flexion control modes to be selected.

- i. Basic mode: Users manually specify the training fingers and flexion angle
- ii. Action mode: Device actively moves and flexes according to the setting procedures.

The device also has two extension control modes:

- i. Controlled deflation: user specifies the speed at which the deflation causes the finger module extension.
- ii. Natural deflation: the electromagnetic valve connecting to the air opens up, causing the fingers to extend quickly in a natural manner.

2) Diagnosis function

- i. Parkinson's rating: The program gives the suggested Parkinson's scale rating based on accelerometer's data.
- ii. The program assesses the recovery status by analyzing the maximum flexion angle changes in the patient's self-training finger movements over a period of time.

3) Life assistance - Assisting mouse click

The device features a "mouse click assistance" mode to prevent inadvertent clicks caused by finger tremors in Parkinson's patients using a computer mouse.

4) Bluetooth communication with Mobile/Pad.

4. Results and Discussion

As stated in chapter 1, the device includes three main features: rehabilitation, diagnosis or classification of the degree of the Parkinson disease and assisting to prevent accidental mouse

clicks. Experiments were conducted to verify their feasibility, effectiveness and accuracy in this chapter.

4.1 Experiment 1: Finger Flexion Angle Accuracy and Repeatability

Objectives:

- 1) Figure out the relationship between the finger flexion degree and the value of Flex sensor.
- 2) Validate the accuracy of the finger flexion angles.
- 3) Validate the analysis of the trend of rehabilitation via the maximum flexion angle that the patient can do intentionally.

Procedures – Objective 1:

- 1) Initiate the Arduino's serial port to observe real-time flex sensor data.
- 2) Manually bend the finger to a series of specific given angles from 15-70 degree.
- 3) After flexion, measure the angle between the line connecting the fingertip to the finger base and the horizontal orientation of the finger.
- 4) Record the output values of the flex sensor at the corresponding finger flexion angles.

Results and Discussions – Objective 1:

The flex sensor data, paired with the actual finger flexion angles, are represented graphically in Figure 4.1.

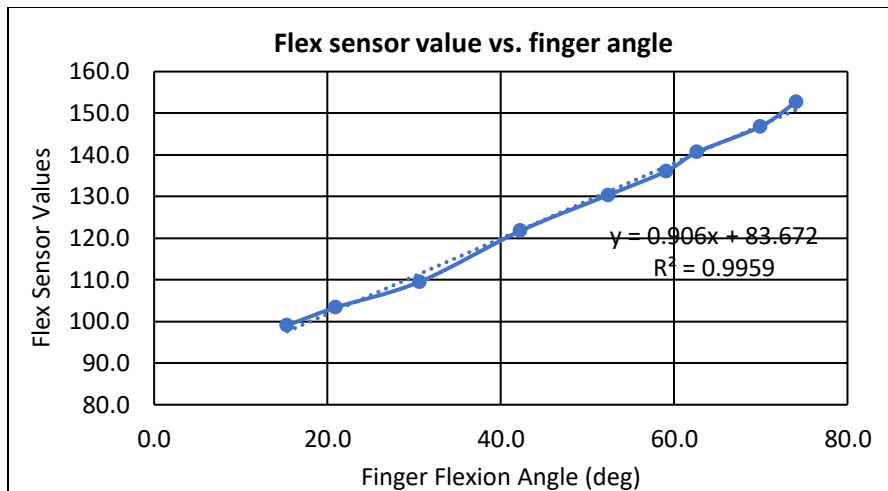


Figure 4.1: Flex sensor values in relation to finger actual flexion angles

According to Figure 6.1, the actual finger bend angle and the data from the bend sensor roughly exhibit a linear relationship. An equation that can provide a rough conversion between the two has been derived: $y = 0.906x + 83.672$, where y represents the bend sensor output value, and x represents the actual finger bend angle. The R-squared value for both is 0.9959, indicating relatively reliable data. 83.672 is the Flex sensor value when the Flex sensor is placed horizontally with no flexion.

This relationship is used to control the flexion angle subsequently in the following procedures.

Procedure – Objective 2 & 3:

- 1) Initiate the Arduino's serial port to observe real-time flex sensor data.
- 2) Run the program from laptop or mobile phone.
- 3) Enter the desired finger flexion angle, starting from 10 degrees and increasing in 10-degree increments up to 70 degrees.
- 4) After each flexion, record the angle between the straight line from the fingertip to the finger base and the finger's horizontal plane.
- 5) Perform the procedure 10 times.

6) Calculate and analyze the accuracy of the angles.

Results and Discussions – Objective 2 & 3:

The deviation of Flex sensor data at various finger angles, obtained through 10 repeated tests, is illustrated in Figure 4.2.

The absolute deviation between actual finger angle and preset finger angle ranges between 0.4 and 4.9 degree, and the relative deviate ranges between 2.1-7.6%. Both the absolute and relative deviation increases in trend with the flexion angle increases as shown in Figure 4.2.

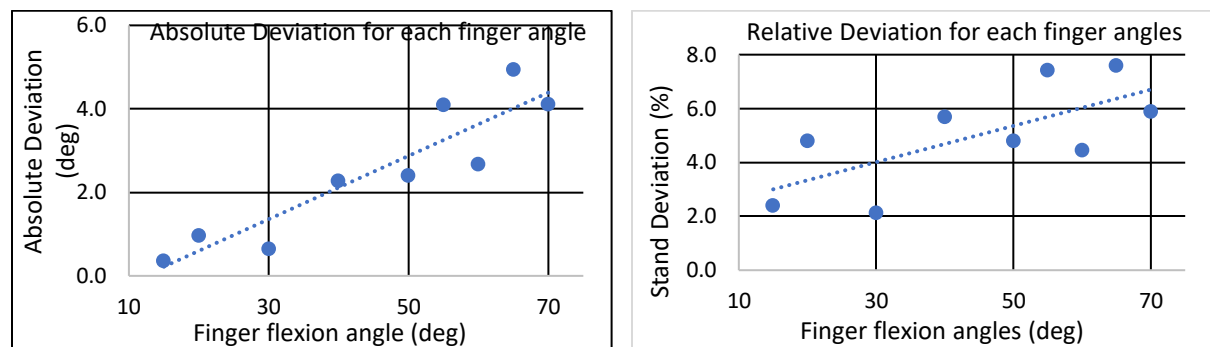


Figure 4.2: Flex sensor values in relation to finger actual flexion angles

The standard deviation of the 10 times repeated test data for each finger angle is shown in Figure 4.3, which ranges from 0.2-3.7 degree.

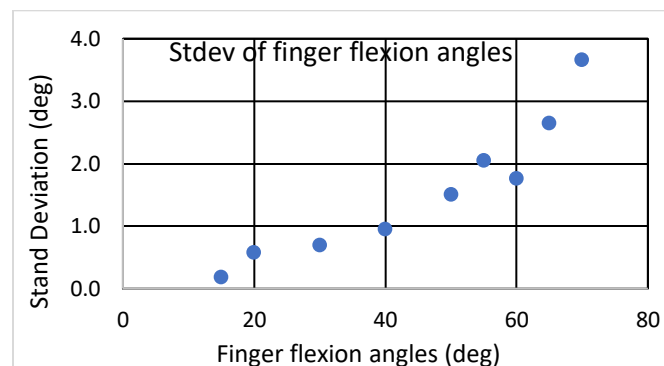


Figure 4.3: standard deviation of the 10 times repeated test data for each finger flexion angle

Validate the analysis of the trend of rehabilitation:

The rehabilitation status is assessed by tracking the changes in the maximum angle to which the patient can intentionally flex their fingers over time. This maximum angle is simulated across a time period and depicted in Figure 4.4.

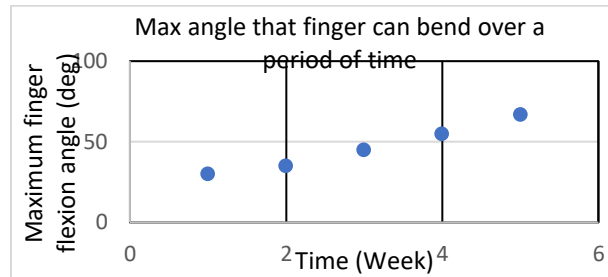


Figure 4.4 Data analysis for trend of rehabilitation

Conclusion:

The flex sensor value demonstrates a strong linear correlation with the finger flexion angle, allowing for precise control of finger movements. The device's flexion range satisfies clinical requirements with a maximum of 70 degrees. The relative deviation of the flexion angle from the set value ranges between 2.1% and 7.6% within this range, ensuring an accuracy better than $\pm 10\%$. Monitoring the trend of the maximum angle to which the fingers can flex over time provides insights into the rate of rehabilitation progress.

4.2 Experiment 2: Diagnosis/rating Parkinson patient with finger tremor

The movement disorder is very common for Parkinson patients. There are various methods to rate the scale of the movement disorder ^[18]. This device deploys the “Constancy of Rest Tremor” method to rate the scale of Parkinson to provide diagnosis suggestions.

Objective:

Rate the scale of Parkinson by using “Constancy of Rest Tremor”.

Procedure:

- 1) Initiate and set the Arduino serial port BAUD rate to 750 to observe real-time data from the Accelerometer attached to the back of the hand.
- 2) Keep hand still without movement. Get the data of the accelerometer for 10 seconds to set the threshold of tremor (silent output).
- 3) Simulate the situation where the handshakes constantly and record the data from the accelerometer for 10 seconds to figure out which axis (es) of the accelerometer has or have change that's significant enough so that it could be used in determining tremor cases.
- 4) Simulate the situation where hand's having tremor occasionally and control the tremor and stillness so that they are about 50% each over a period of time.
- 5) Keep track of the data for another 3 minutes and record all the data from the Accelerometer.
- 6) Compare the data values with the threshold established in step 2 to calculate the percentage of data points that exceed the threshold relative to the total data set.

The rating criteria are below according to reference ^[18]:

- a) Normal: No tremor
- b) Slight: Tremor at rest is present $\leq 25\%$ of the entire examination period.
- c) Mild: Tremor at rest is present 25-50% of the entire examination period.
- d) Moderate: Tremor at rest is present 51-75% of the entire examination period.
- e) Severe: Tremor at rest is present $\geq 75\%$ of the entire examination period.

Results and Discussions:

The values from the accelerometer, recorded over a 10-second period while the hand remains still, are displayed in Figure 4.5.

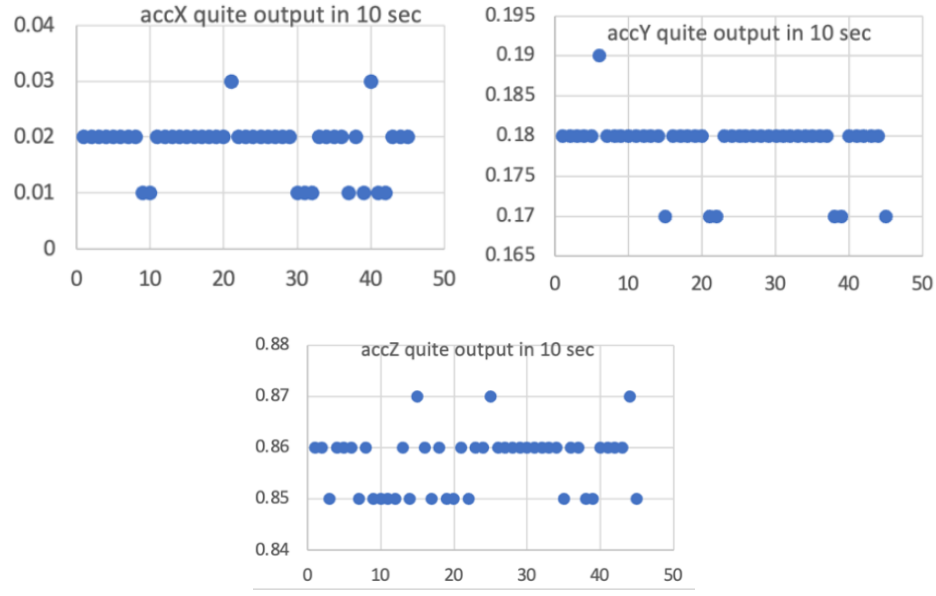


Figure 4.5: accelerometer output (X, Y, Z) over a 10-second period while the hand remains still

For each axis, we calculated their average and standard deviation, and threshold which are expressed as: average $\pm$ standard deviation. They are summarized in Table 4.1.

Table 4.1: accelerometer output (X, Y, Z) over a 10-second period while the hand remains still

	accX	accY	accZ
Average	0.018444	0.178889	0.857333
STDEV	0.004746	0.003827	0.0058
Threshold	Average $\pm$ STDEV		

The data simulating constant tremors over a 10-second period are compared with thresholds for the X, Y, Z axes from the accelerometer and presented in Figure 4.6.

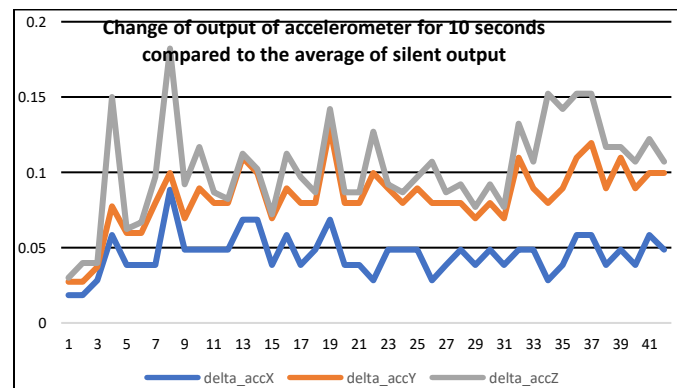
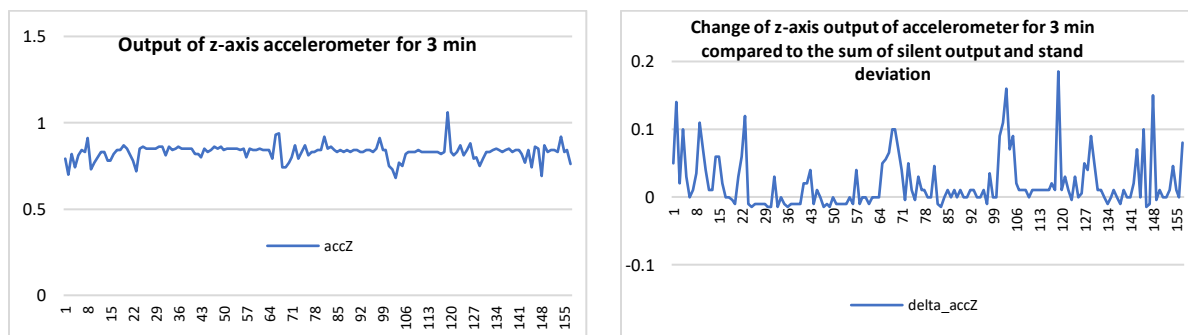


Figure 4.6: output of accelerometer (X, Y, Z) for simulated constant hand tremor over a 10-second period

Figure 4.6 illustrates that the z-axis values exhibit the most significant changes during finger/hand tremors, indicating higher sensitivity to movement. Consequently, the z-axis values are utilized for tremor evaluation. The accelerometer output changes in the Y and Z directions are more pronounced, making either axis a potential candidate for assessing tremor severity. Data from the accelerometer's z-axis, collected over three minutes while simulating occasional finger/hand tremors, are displayed in Figure 4.7.



a) Output of the z-axis accelerometer

b) Change of output of accelerometer z-axis

Figure 4.7: output of accelerometer z-axis for 3 min for simulated finger/hand tremor. Data are compared against established thresholds: readings below the lower threshold or above the upper threshold are classified as "Tremor"; otherwise, they are considered "Not Tremor." In Figure 4.7(b), any value greater than 0 is categorized as "Tremor."

During this simulation, 84 out of 157 readings (54%) exceed the threshold, indicating a "Moderate" tremor rating, aligning with the simulation of approximately 50% tremor occurrence.

Conclusion:

The accelerometer's z-axis sensitivity makes it suitable for Parkinson's tremor rating when attached to the back of the hand. However, values from other axes might also be utilized,

depending on the accelerometer's position and orientation on the hand. The tremor rating results from the shaking simulations suggest a high degree of accuracy. Top of Form Bottom of Form

4.3 Experiment 3: life assistance -assisting mouse clicks for Parkinson patient

While using a mouse, a Parkinson's patient may inadvertently click due to finger tremors before the cursor reaches the intended target. This device features a "life assistance" mode to prevent such accidental clicks when the hand is in motion towards the target.

Objective:

Verify that the device, equipped with an accelerometer, does not initiate finger flexion to click the mouse unless the hand has stopped moving towards the target for more than 2 seconds.

Procedure:

- 1) Activate the device's "life assistance" mode. Grasp the mouse with hand.
- 2) Maintain the mouse in a stationary position for 20 seconds and record the accelerometer data (x, y, z axes) as the reference value for setting the threshold.
- 3) Navigate the mouse cursor on the screen continuously for 1 minute without stopping.
Record the accelerometer data. Send a command to attempt flexing the index finger to click the mouse and verify that the index finger does not flex
- 4) Continuously navigate the mouse cursor for 1 minute, attempting to flex the index finger to click. Verify the index finger does not flex. Record the accelerometer data.
- 5) Cease hand movement for 2 seconds. Send a command on laptop or mobile phone to attempt flexing the index finger to click the mouse and verify that the index finger does flex.

Results and Discussions:

The x, y, z values of the accelerometer when the mouse remains still are listed in Table 4.2, which includes the average and standard deviation for the data collected over 20 seconds. The threshold is defined as the average $\pm$ standard deviation.

Table 4.2: accelerometer output (X, Y, Z) when the mouse is stationary for 20 seconds

	accX	accY	accZ
Average	-0.02425	0.00725	0.8925
STDEV	0.00813	0.00599	0.008086
Threshold	Average $\pm$ STDEV		

Figure 4.8 shows the changes in accelerometer output along the X, Y, and Z axes over a 20-second period during mouse movement. The more substantial changes in the Y and Z axes, as compared to the X axis, indicate that these are the sensitive directions for mouse movement. Thus, either the Y or Z axis will be utilized for the monitoring of mouse movement in subsequent procedures.

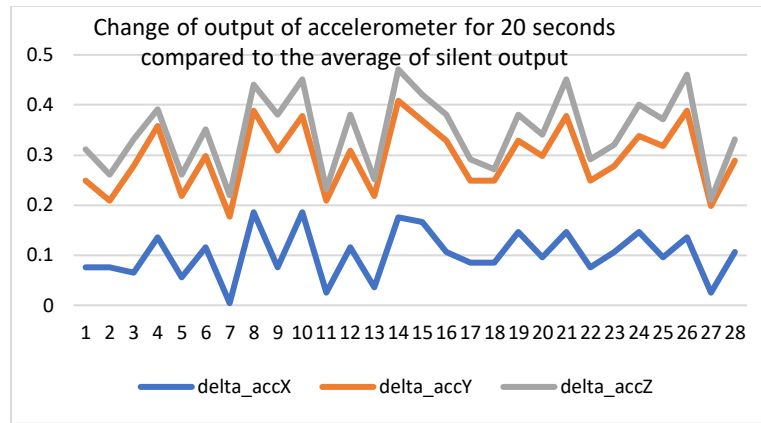


Figure 4.8: Change of accelerometer output (x, y, z) for 20 seconds for mouse movement

The accelerometer data change during continuous mouse movement relative to the stationary state is shown in Figure 4.9.

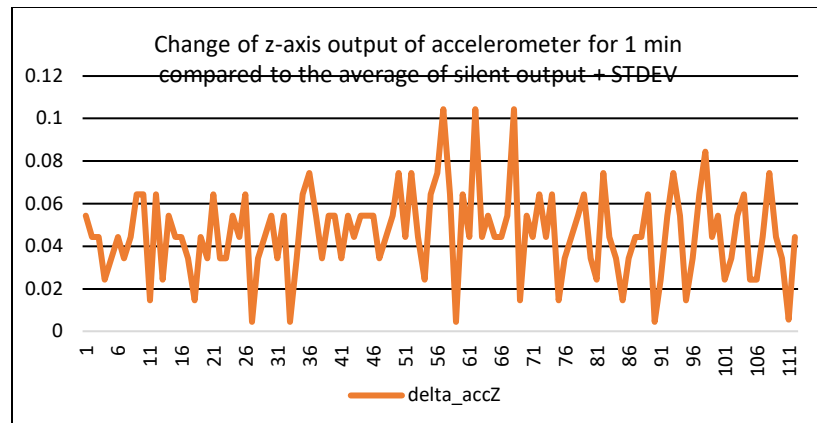


Figure 4.9: The data of accelerometer for continuous movement for 1 minute

An attempt to activate the device with a 'start' command on laptop or button on mobile phone while the mouse is in continuous motion shows that the device cannot be activated. However, once the mouse remains stationary for 2 seconds and a 'start' command is issued again, the device successfully activates and facilitates the bending of the index finger.

The device's software is capable of disabling unintentional finger movements during hand motion and enabling finger movement after the hand has been still for more than 2 seconds, which may assist patients with Parkinson's disease in effectively operating the mouse.

5. Conclusions and Outlooks

Based on the design and tests performed above, the device can achieve the designed features and functionalities which meet the intended user needs very well.

Users are provided with a new feature of operating the device using a newly developed mobile app, proving users with higher convenience.

The project summary is as follows:

Finger flexion:

- 1) The finger module can achieve precise control, with accuracy better than $\pm 10\%$ within the designed angle range of up to 70 degrees.

- 2) The trend of the maximum flexion angle over time can indicate the progression of rehabilitation offline.
- 3) The finger module offers high flexibility and user-friendly.

Scale rating of finger/hand tremor for Parkinson's:

Utilizing the 'Constancy of Rest Tremor' metric, the device effectively provides a scaled rating of finger tremor for Parkinson's patients.

Assisting mouse clicks for Parkinson's patient:

The device can inhibit finger movement to prevent accidental finger bending during hand motion and enables finger movement after the hand remains still for a duration exceeding 2 seconds, assisting Parkinson's patients in operating the mouse effectively.

During the testing process, several areas for potential improvement have been identified for future development:

- 1) Enhancing the comfort and convenience of the Rehabilitation device.
- 2) Integrating the functionality for analyzing the progression of rehabilitation of Parkinson's patients' hand and more data analysis features in the software.
- 3) Applying the device to more patients' hands to collect more real-world data and verify its effectiveness.

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